# Redlich–Kwong equation of state

The Redlich–Kwong equation of state is a two-constant cubic equation:

$$p = \frac{RT}{V_\mathrm{m} - b} - \frac{a}{T^{1/2} V_\mathrm{m}(V_\mathrm{m} + b)}$$

- $p$ is pressure
- $T$ is the absolute temperature
- $R$ is the gas constant
- $V_\mathrm{m}$ is the molar volume
- $a$ is a constant reflecting attraction between molecules
- $b$ is the excluded volume of molecules

Constants $a, b$ depend on the critical temperature and pressure via equations:

$$a = \frac{1}{9(2^{1/3} - 1)} \frac{R^2 T_\mathrm{c}^{5/2}}{p_\mathrm{c}}, \qquad b = \frac{2^{1/3} - 1}{3} \frac{R T_\mathrm{c}}{p_\mathrm{c}}$$

- Actually, $a, b$ are back-calculated from the experimental $p_\mathrm{c}$, $T_\mathrm{c}$ to use the equation!

# Quality



- accuracy $\sim 2$ to 5%
- simple
- the critical compressibility factor

$$z_\mathrm{c} = \frac{p_\mathrm{c} V_\mathrm{c}}{R T_\mathrm{c}} = \frac{1}{3}$$

  is much better than for the van der Waals equation ($z_\mathrm{c} = 3/8$)
  for most gases (excl. water vapor): $z_\mathrm{c} \approx 0.28$–$0.30$
- bad near the critical point: no classical mean-field equation of state can handle it

# Volume



The Redlich–Kwong equation is cubic in volume. Thus, it has up to three roots:

- one real root for $T > T_c$
- a real triple root for $T = T_c$ (singular case)
- up to three real roots for $T < T_c$:
  – liquid (stable or metastable)
  – vapor (stable or metastable)
  – unstable phase in between (immediate spinodal decomposition)
  – if there are 2 or more roots, the stable root has the lowest fugacity
- The JavaScript applet calculator uses the secant method to find the roots.

# Fugacity coefficient



- The fugacity coefficient

$$\ln\varphi = \int_{V_m(p)}^{\infty}\left[\frac{p(V_m)}{RT}-\frac{1}{V_m}\right]\mathrm{d}V_m - \ln z + z - 1 = \int_0^{c(p)}\left[\frac{p(1/c)}{RT}-c\right]\frac{\mathrm{d}c}{c^2} - \ln z + z - 1$$

Substitution $c = 1/V_m$ is suitable for numerical integration.

- For Redlich–Kwong, the integration can be performed analytically:

$$p = p(V_m, T) = \frac{RT}{V_m - b} - \frac{a}{\sqrt{T}V_m(V_m + b)}$$

$$z = \frac{V_m}{V_m - b} - \frac{a}{RT^{3/2}(V_m + b)},$$

$$\ln\varphi = \ln\frac{V_m}{V_m - b} + \frac{a}{RT^{3/2}b}\ln\frac{V_m}{V_m + b} - \ln z + z - 1$$

- The fugacity $f = \varphi p$ is used to determine stability for $T < T_c$